

Bitwise linear trail through **QALQAN** found by SAT

CiVerLy

August 21, 2026

1 QALQAN

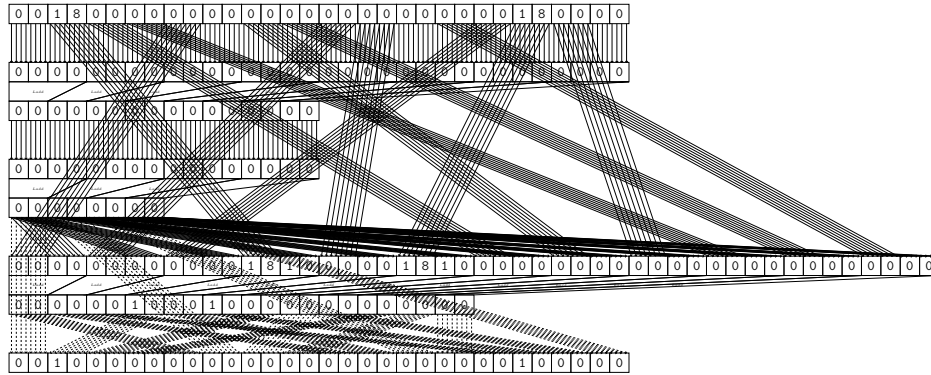
Maximal linear correlation: 2^{-58} [illegible]

2 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

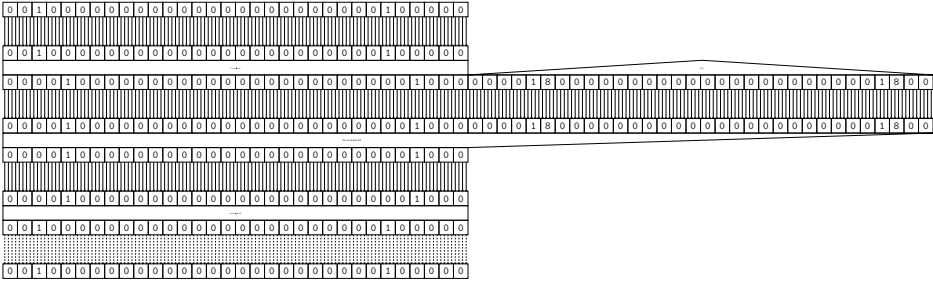
0	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	0	0	0	0
0	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	f	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0	0	0
0	0	1	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	8	0	0	0	0

3 L

Maximal linear correlation: $2^{-2.0}$ 

4 Add128

Maximal linear correlation: $2^{-2.0}$

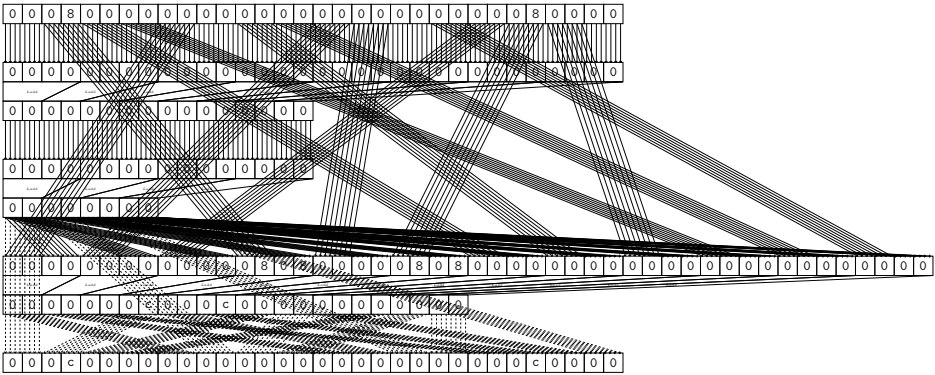


5 SBoxLayer

Maximal linear correlation: $2^{-6.0}$ [illegible]

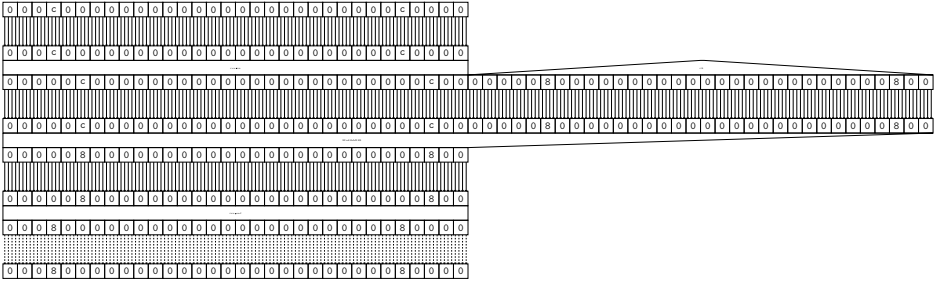
6 L

Maximal linear correlation: $2^{-2.0}$



7 Add128

Maximal linear correlation: $2^{-2.0}$



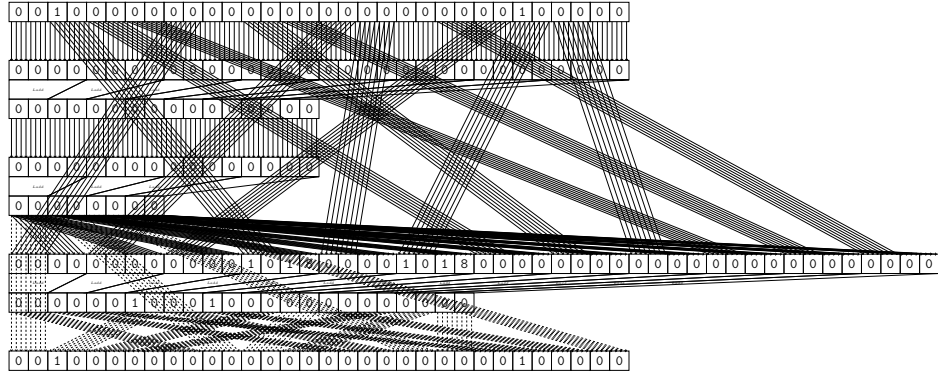
8 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0
0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
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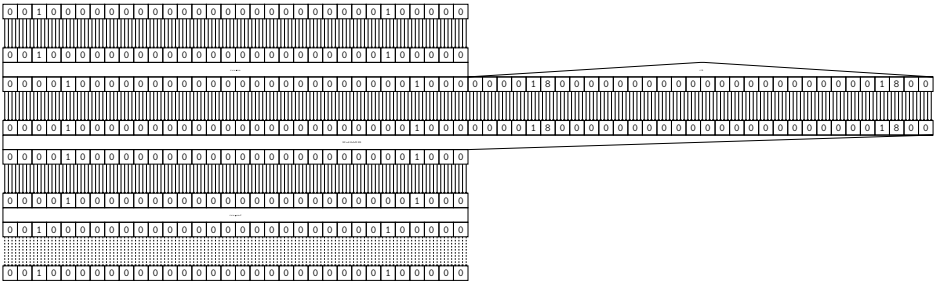
9 L

Maximal linear correlation: $2^{-2.0}$



10 Add128

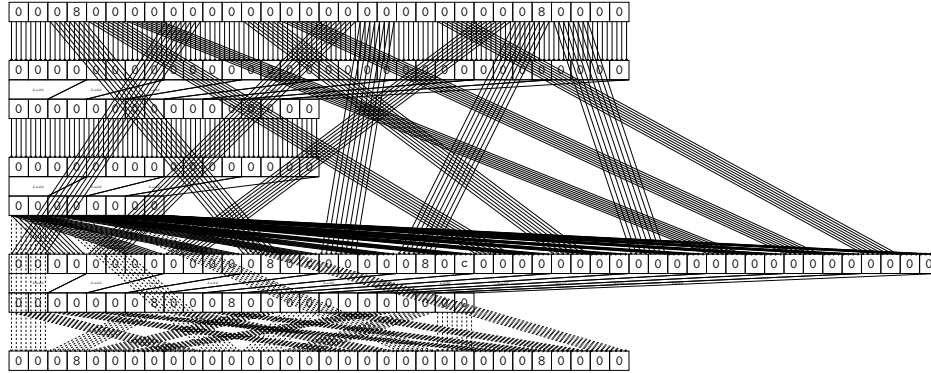
Maximal linear correlation: $2^{-2.0}$



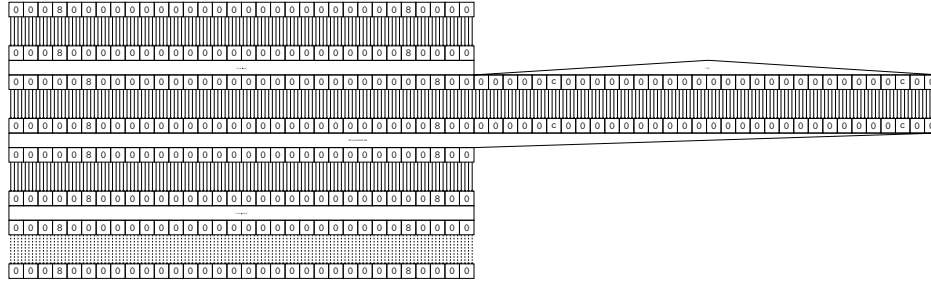
11 SBoxLayer

Maximal linear correlation: $2^{-6.0}$ [illegible]

12 L

Maximal linear correlation: $2^{-2.0}$ 

13 Add128

Maximal linear correlation: $2^{-2.0}$ 

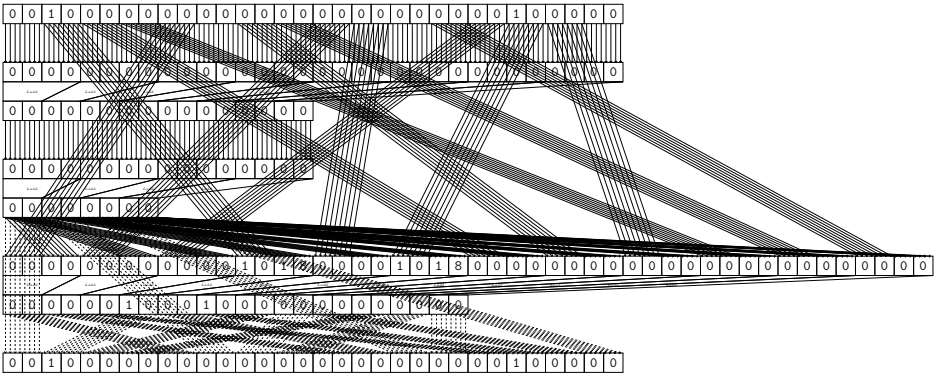
14 SBoxLayer

Maximal linear correlation: $2^{-6.0}$

0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0
0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

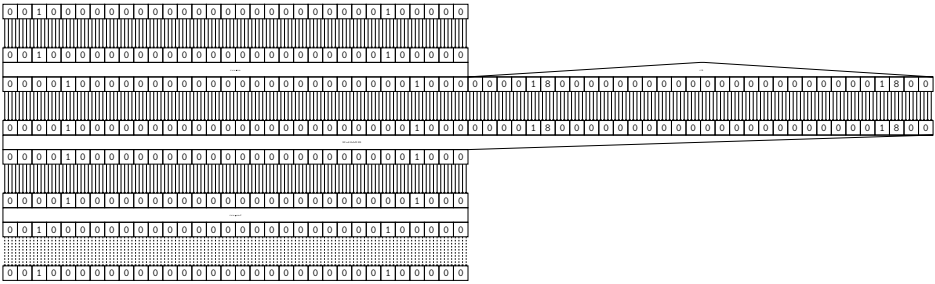
15 L

Maximal linear correlation: $2^{-2.0}$



16 Add128

Maximal linear correlation: $2^{-2.0}$



17 SBoxLayer

Maximal linear correlation: $2^{-6.0}$ [illegible]

18 L

Maximal linear correlation: $2^{-2.0}$

